

Project Initialization and Planning Phase

Date	29 September 2026
Team ID	SWTID-2026-7019
Project Title	EduGenie - AI Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform self-study using generative AI, making learning faster, simpler and more accessible. It tackles scattered, costly and hard-to-use learning tools, promising quicker doubt resolution, better revision and more confident students. Key features include an AI Q&A assistant, simple explanations, summaries, quizzes and learning paths.

Project Overview	
Objective	The primary objective is to help students learn faster and with more confidence by giving them instant answers, simple explanations, summaries, practice quizzes and personalised learning paths from a single AI-powered web application.
Scope	EduGenie is a web application with five AI modules - Q&A, Explanation, Summarization, Quiz Generation and Learning Path - exposed as REST endpoints through FastAPI. Text is processed by Google Gemini (google-genai), with a local Flan-T5 model as a fallback for explanations. User accounts, progress tracking and a mobile app are outside the current scope.
Problem Statement	
Description	Students often have doubts outside class hours, struggle with long or complex study material, and have no simple way to test themselves or decide what to learn next. Existing tools are scattered, paid or not designed for learners, so studying becomes slow and stressful.
Impact	Solving this gives learners on-demand, low-cost academic support: faster doubt resolution, better understanding and revision, regular self-testing and a clear study roadmap, leading to higher confidence and better learning outcomes.
Proposed Solution	
Approach	Use a generative AI model with task-specific prompts to answer questions, explain concepts at school level, summarise text, produce multiple-choice quizzes in JSON and build adaptive, level-wise learning paths, all delivered in real time through a lightweight web interface.

Key Features	- AI Q&A for any academic doubt. - Simple, school-level concept explanations. - Plain-language text summarization. - Auto-generated 3-question MCQ quizzes with instant answer feedback. - Beginner/intermediate/advanced learning path recommendations. - Real-time responses; model and API key configurable through environment settings.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	4-core CPU; GPU (e.g. T4) optional for the local Flan-T5 model
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	10 GB SSD (about 3 GB for the cached local model)
Software		
Frameworks	Python frameworks	FastAPI, Uvicorn, Jinja2
Libraries	Additional libraries	google-genai, transformers, torch, sentencepiece, python-dotenv, python-multipart
Development Environment	IDE	Python 3.13 (venv), VS Code with Live Server
Data		
Data	Source, size, format	No training dataset: pretrained Google Gemini and MBZUAI/LaMini-Flan-T5-783M; user-entered text; JSON over REST